GATE-AII BRANCHES
Engineering Mathematics



SINGLE VARIABLE CALCULUS

Lecture No.- 12



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Recap of previous lecture



Topic Integration via parts

Topic Definite integrals

Topic Improper integrals

Topic Area bounded regions

Topics to be covered



Topic Concept of Taylors series

Topic Problems based on Taylors series

Topic Concept of partial derivatives

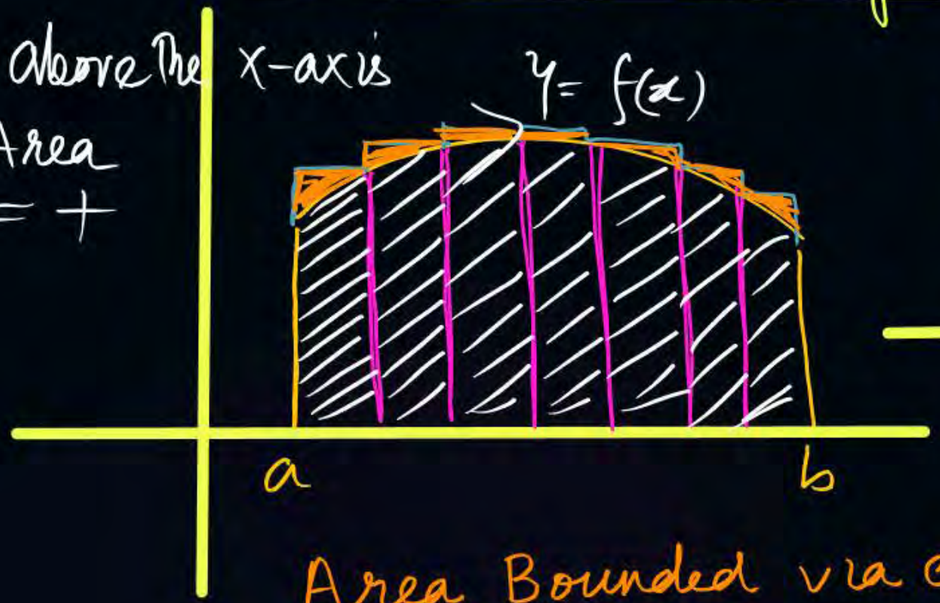
Topic Problems based on partial derivatives

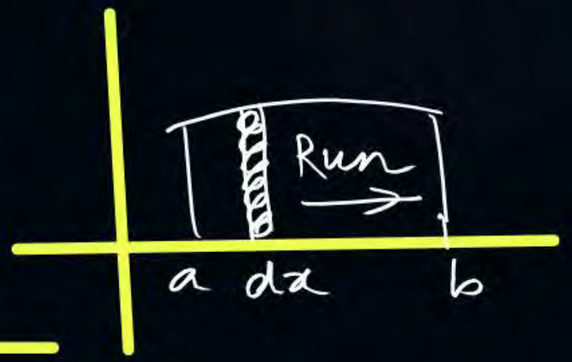
Area bounded via regions

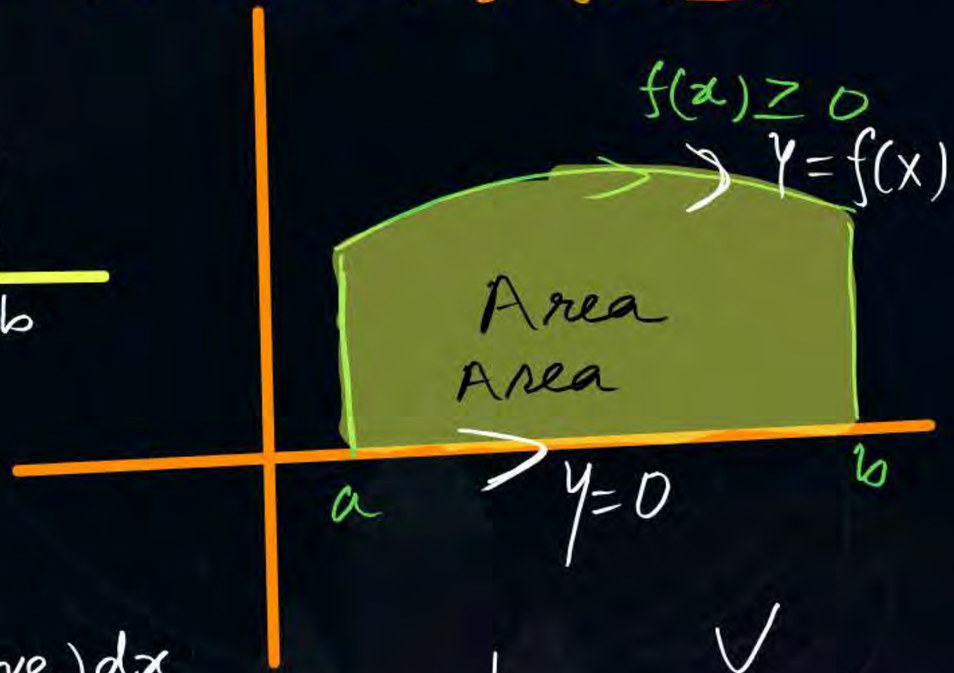
$y = f(x) \; \forall \; x \in [a,b] \quad f(x) \geq 0$

Above the x-axis

Area = +



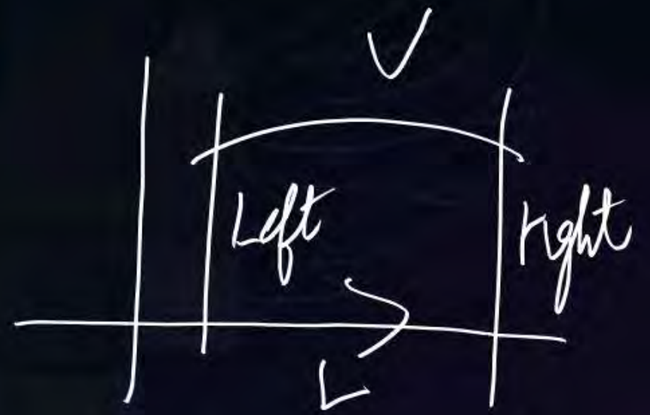




Area Bounded via curve

$$\text{AREA } A = \int_a^b (\text{upper curve} - \text{lower curve})\,dx$$

$$= \int_a^b [f(x) - 0]\,dx = \int_a^b f(x)\,dx = \int_a^b y\,dx$$

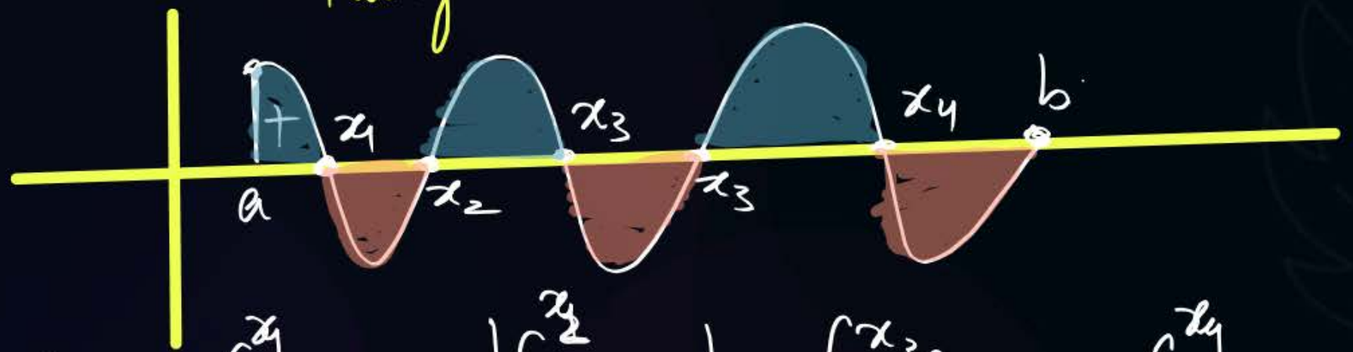


CASE 02 If $y = f(x)\ \forall\ x \in [a,b]$ $f(x) \le 0$

$$\text{Area } A = \left| \int_a^b f(x)\,dx \right| \rightarrow \text{below the } x\text{-axis}$$



CASE 03 If $y = f(x)$ crosses the Horizontal axis



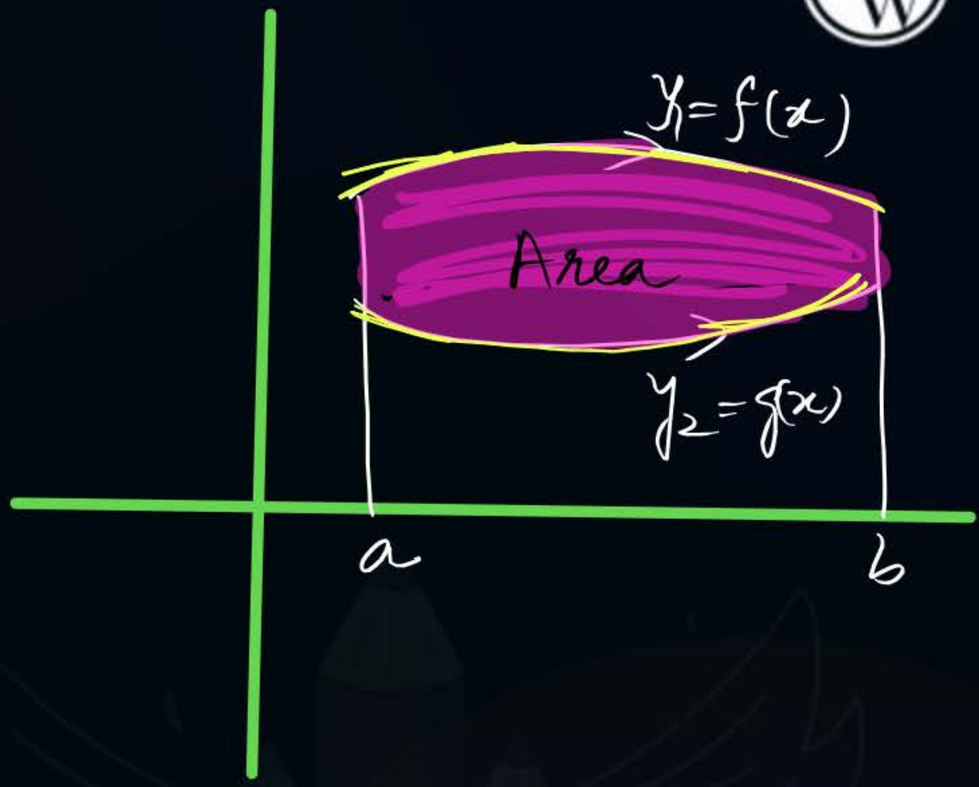
$$\text{Area} = \int_a^{x_1} f(x)\,dx + \left| \int_{x_1}^{x_2} f(x)\,dx \right| + \int_{x_2}^{x_3} f(x)\,dx + \int_{x_3}^{x_4} f(x)\,dx + \left| \int_{x_4}^{b} f(x)\,dx \right|$$

CASE 04 If Two curves Are given

$y_1 = f(x)$ $y_2 = g(x)$

Area Bounded via curve

$$A = \int_a^b [f(x) - g(x)]\, dx$$





Topic : Single Variable Calculus

#Q. The area bounded by the curve $y = x^2$ and $x^2 + y^2 = 2$ above X-axis is

A $\frac{1}{6} + \frac{\pi}{4}$

B $\frac{\pi}{4} + \frac{1}{6}$

C $\frac{2}{3} + \frac{\pi}{2}$

D $\frac{1}{3} + \frac{\pi}{2}$

STEP 01 Plot The Curves $C_1 : y = x^2$ — Parabola $y = x^2$, $x^2 + y^2 = 2$

$C_2 : x^2 + y^2 = 2$ circle $y = \sqrt{2 - x^2}$

Step 02 Calculate The Points of Intersection

$x^2 + y^2 = 2$, $y = x^2$

$x^2 + x^4 - 2 = 0$ biquadratic equⁿ

$x^2 = t$

$t + t^2 - 2 = 0$

$t^2 + t - 2 = 0$

$t^2 + 2t - t - 2 = 0$

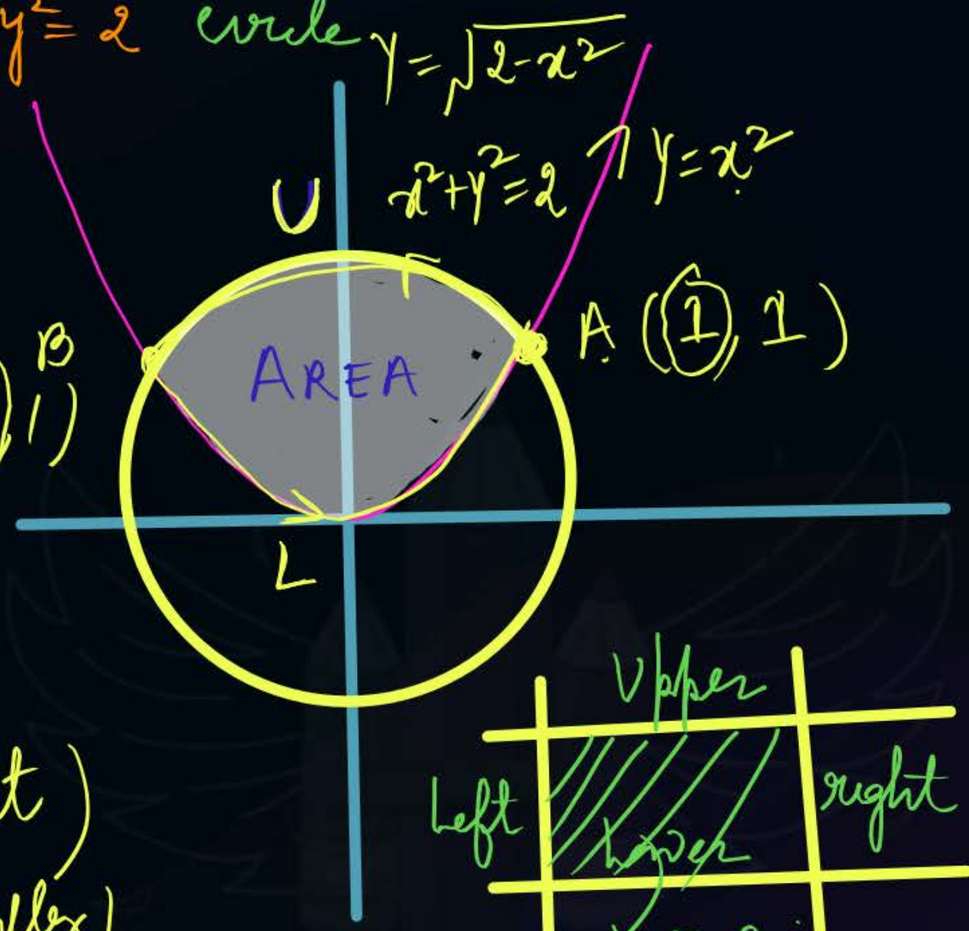
$t(t+2) - 1(t+2) = 0$

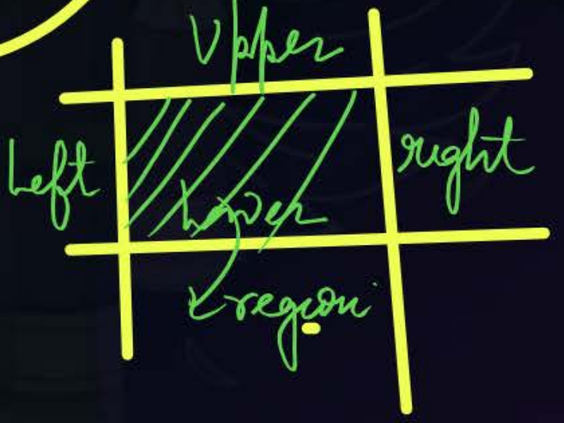
$t = 1$, $t = -2$

$x^2 = 1$

$x = \pm 1$

$x^2 = -2$ (reject) (Complex)





$$\text{Area} = \int_{-1}^{1(\text{right})} \left(\sqrt{2-x^2} - x^2\right) dx$$

Left

$$= \frac{\pi}{2} + \frac{1}{3}$$ Ans.

$x^2+y^2=2$

$y=\sqrt{2-x^2}$



Topic : Single Variable Calculus

12th

MSQ

#Q. Area of the region bounded by the curve $y = e^x$ and lines $x = 0$ and $y = e$ is

A $e - 1$

B $\int_1^e \ln(e + 1 - y)dy$

C $e - \int_0^1 e^x dx$

D $\int_1^e \ln y dy$

$$\text{Area} = \int_0^1 e - e^x = e(1-0) - \int_0^1 e^x\,dx$$

(C)

$$= e - \int_0^1 e^x\,dx$$

$$\text{Area} = \int_1^e (\ln y - 0)\,dy$$

$$= \int_{\text{Lower}}^{\text{upper}} (\text{Right} - \text{Left})\,dy$$

(D)

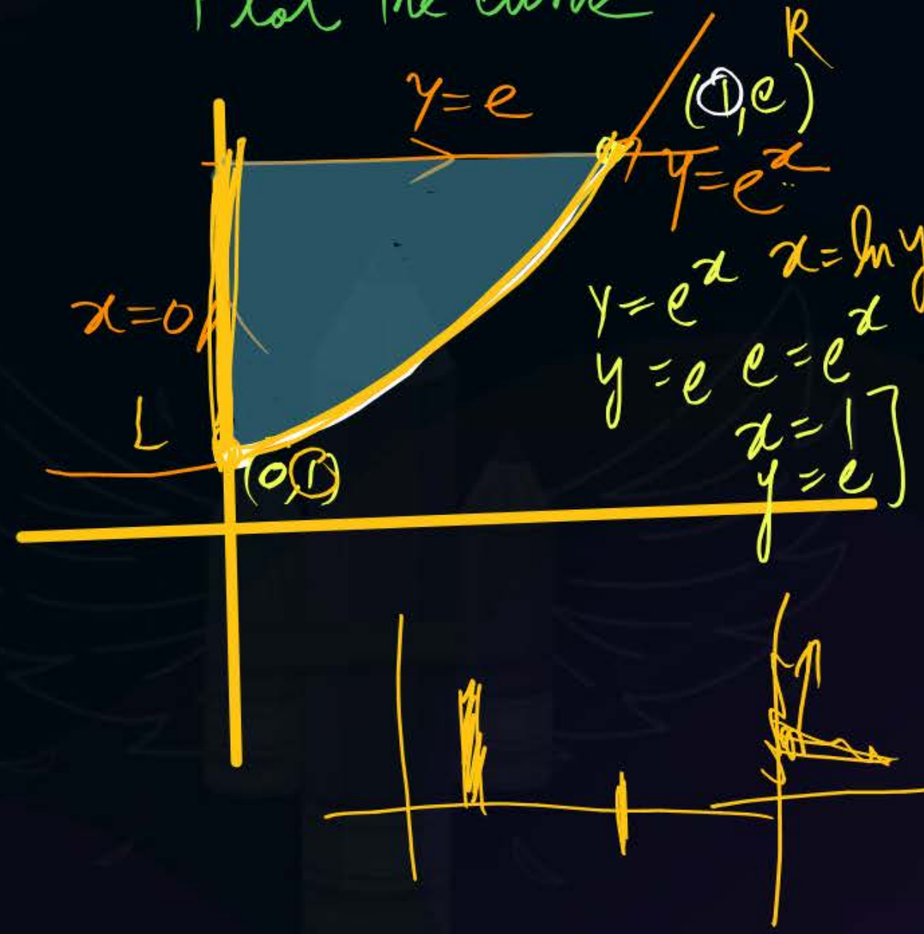
$$\boxed{A = \int_1^e \ln y\,dy}$$

$y = e^x \quad x = 0$

$y = e$

STEP 01

Plot The curve



$$A = \int_1^e \ln y \, dy$$

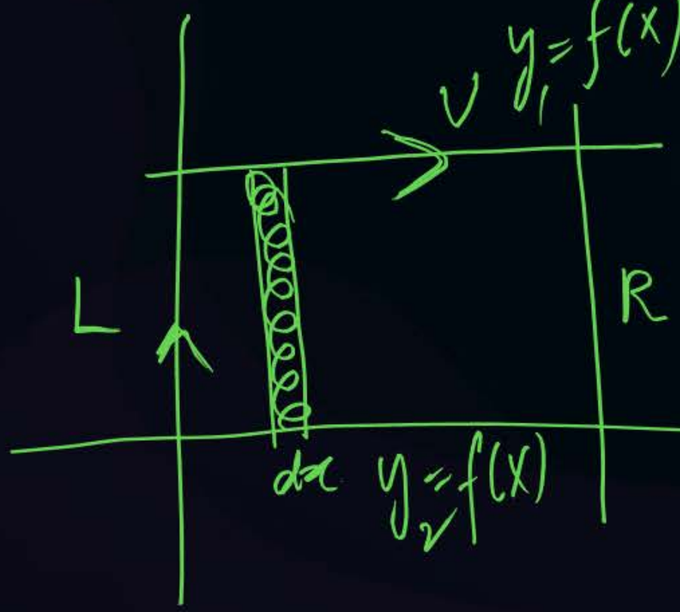
Using Kings Rule

$$A = \int_1^e \ln(1+e-y) \, dy$$

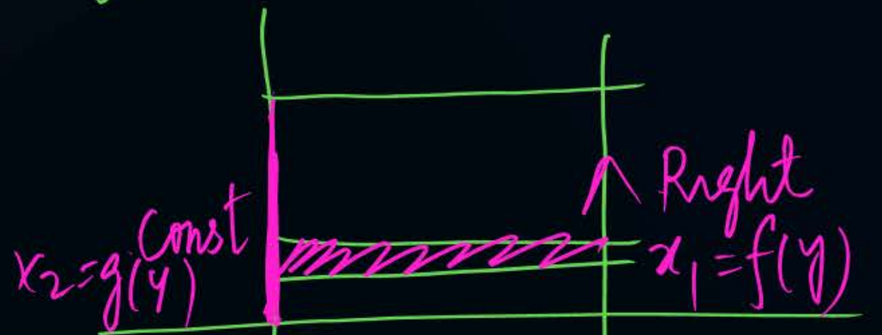
(B) (e) (b)

A X

"If X is constant
Using Vertical strip
y is variable"



$$\int_a^b (y_1 - y_2) \, dx$$



$$\int_{\text{Lower}}^{\text{upper}} (R-L) \, dy$$

Horizontal strip
y constant
x variable

Taylor SERIES

A) Maclaurins SERIES

Increasing

$$f(x) = A_0 + A_1 x + A_2 x^2 + A_3 x^3 + - + A_n x^n$$

Diff. w.r.t to x

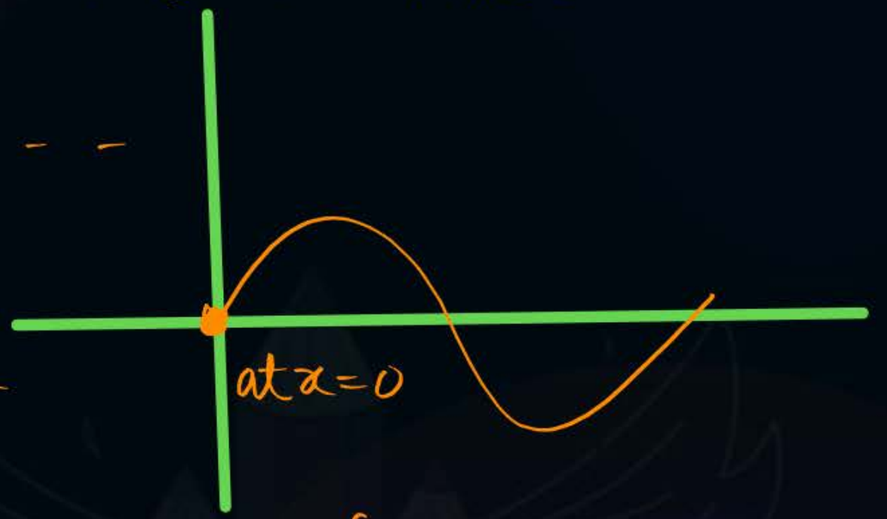
$$f'(x) = 0 + A_1 + 2A_2 x + 3A_3 x^2 + - - -$$

Again Diff.

$$f''(x) = 0 + 2A_2 + 6A_3 x + - - -$$

Again Diff.

$$f'''(x) = 0 + 6A_3 + - - - -$$



At $x=0$

$$f(0) = A_0 \quad \frac{f'(0)}{1!} = A_1 \quad \frac{f''(0)}{2!} = A_2 \quad \frac{f'''(0)}{3!} = A_3$$

$f(x) = \sin x$

$f(x) = e^x$

$$f(x) = f(0) + \frac{x}{1!}f'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \frac{x^4}{4!}f^{iv}(0) + \cdots$$

put the value of constants

$f(x) = e^x$

$f(0) = e^0 = 1$

$f'(0) = e^0 = 1$

$f''(0) = e^0 = 1$

$f'''(0) = e^0 = 1$

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$$ at $x=0$

(2) $f(x) = \sin x$

$$= \sin 0 + \frac{x}{1!}\cos 0 + \frac{x^2}{2!}(-\sin 0) + \frac{x^3}{3!}(-\cos 0) + \frac{x^4}{4!}(+\sin 0)$$

$$= 0 + \frac{x}{1!} + \frac{x^3}{3!}(-1) + 0 - - -$$

$$= x - \frac{x^3}{3!} + - - - -$$

$f(x) = e^{-x}$
$= \cos x$
] SERIES

at $x = 0$

Maclaurins Series

Taylor SERIES

Shift at $x = a$ Point

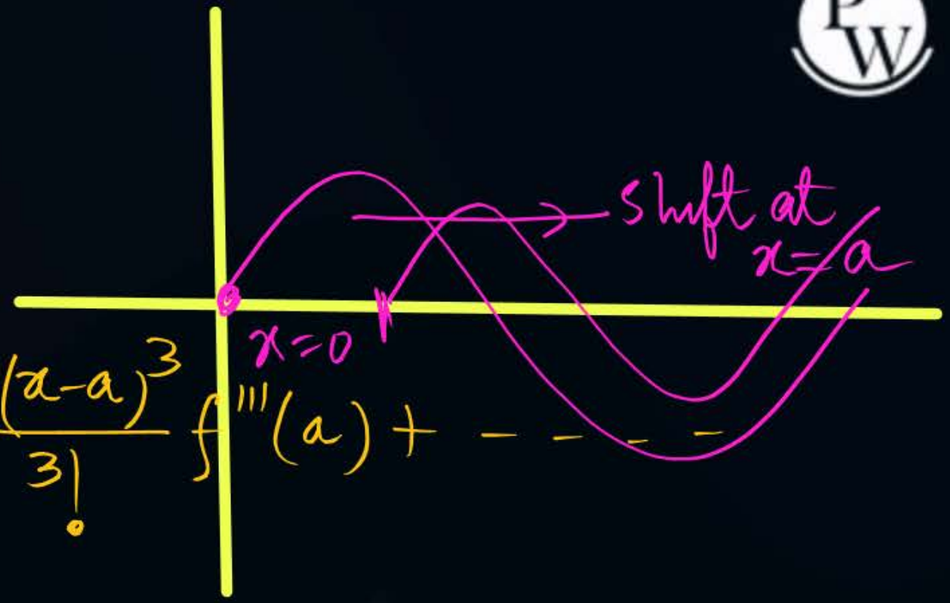
$$f(x) = \underbrace{f(a) + (x-a)f'(a)}_{\text{Linear Term}} + \frac{(x-a)^2}{2!}f''(a) + \frac{(x-a)^3}{3!}f'''(a) + \cdots$$

(linear Approximation)

$$\boxed{f(x) = f(a) + (x-a)f'(a)} \longrightarrow \text{linear Approximation}$$

$$\lim_{x \to a} \frac{f(x) - f(a)}{x-a} = f'(a)$$

= Tangent



Taylor $f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \frac{(x-a)^3}{3!}f'''(a) + \cdots$

If $a = 0$

$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots$

Maclurins SERIES
X=0



Topic : Single Variable Calculus

#Q. The third term in the Taylor's series expansion of e^x about 'a' would be ____.

A $e^a(x-a)$

B $\frac{e^a}{2}(x-a)^2$

C $\frac{e^a}{2}$

D $\frac{e^a}{6}(x-a)^3$

$f(x) = e^x$
$f'(x) = e^x$
$f''(x) = e^x$
$f''(a) = e^a$

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \dots$$

$$= \frac{e^a(x-a)^2}{2!}$$



Topic : Single Variable Calculus

$\sin x = \quad x = \frac{\pi}{6}$

#Q. The Taylor series expansion of $\sin x$ about $x = \frac{\pi}{6}$ is given by

A $\frac{1}{2}+\frac{\sqrt{3}}{2}\left(x-\frac{\pi}{6}\right)-\frac{1}{4}\left(x-\frac{\pi}{6}\right)^2-\frac{\sqrt{3}}{12}\left(x-\frac{\pi}{6}\right)^3+\ldots$

B $x-\frac{x^3}{3!}+\frac{x^5}{5!}-\frac{x^7}{7!}+\ldots$

C $x-\frac{\pi}{6}-\frac{\left(x-\frac{\pi}{6}\right)^3}{3!}+\frac{\left(x-\frac{\pi}{6}\right)^5}{5!}-\frac{\left(x-\frac{\pi}{6}\right)^7}{7!}$

D $\frac{1}{2}$

$= f\left(\frac{\pi}{6}\right)+\left(x-\frac{\pi}{6}\right)f'\left(\frac{\pi}{6}\right)+\frac{\left(x-\frac{\pi}{6}\right)^2}{2!}f''\left(\frac{\pi}{6}\right)+\ldots$



Topic : Single Variable Calculus

#Q. For the function e^{-x}, the linear approximation around x = 2 is

$= f(a) + (x-a) f'(a)$

$= f(2) + (x-2) f'(2)$

$= e^{-2} + (x-2)(-e^{-2})$

$= (3-x)e^{-2}$

A $(3 - x)e^{-2}$

B $1 - x$

C $|3 + 2\sqrt{2} - (1 + \sqrt{2})x|e^{-2}$

D e^{-2}



Topic : Single Variable Calculus

$D(e^x + \sin x) = e^x + \cos x$

$D^2(e^x + \sin x) = e^x - \sin x$

#Q. In the Taylor series expansion of $(e^x + \sin x)$ about the point $x = \pi$, the coefficient of $(x - \pi)^2$ is

$e^x + \sin x$

$x = \pi$

coefficient $(x-\pi)^2$

$= f''(a)\dfrac{(x-\pi)^2}{2!} =$ coefficient of $(x-\pi)^2$

$\dfrac{f''(a)}{2!} = \dfrac{(e^\pi - \sin\pi)}{2!} = \dfrac{e^\pi}{2} = 0.5e^\pi$

A. e^π

B. $0.5e^\pi$

C. $e^\pi + 1$

D. $e^\pi - 1$



Topic : Single Variable Calculus

#Q. The Taylor series expansion of $\frac{\sin x}{x-\pi}$ at $x = \pi$ is given by

A $1+\frac{(x-\pi)^2}{3!}+\ldots$

B $-1-\frac{(x-\pi)^2}{3!}+\ldots$

C $1-\frac{(x-\pi)^2}{3!}+\ldots$

D $-1+\frac{(x-\pi)^2}{3!}+\ldots$



Topic : Single Variable Calculus

H.W

#Q. The infinite series $1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\frac{x^4}{4!}+\ldots$ corresponds to

A $\sec x$

B e^x

C $\cos x$

D $1+\sin^2 x$



Topic : Single Variable Calculus

H.W

#Q. The Taylor series expansion of $(3 \sin x + 2 \cos x)$ is

at $x = 0$

A $2 + 3x - x^2 - \frac{x^3}{2} + \ldots$

B $2 - 3x + x^2 - \frac{x^3}{2} + \ldots$

C $2 + 3x + x^2 + \frac{x^3}{2} + \ldots$

D $2 - 3x - x^2 + \frac{x^3}{2} + \ldots$



Topic : Single Variable Calculus

H.W

#Q. Let $f(x) = e^{x+x^2}$ for real x. From among the following, choose the Taylor series approximation of f(x) around x = 0, which includes all powers of x less than or equal to 3.

A $1 + x + x^2 + x^3$

B $1 + x + \frac{3}{2}x^2 + x^3$

C $1 + x + \frac{3}{2}x^2 + \frac{7}{6}x^3$

D $1 + x + 3x^2 + 7x^3$



Topic : Single Variable Calculus

CSE - Optional Maths

#Q. Taylor series expansion of $f(x)=\int_0^x e^{-\left(\frac{t^2}{2}\right)}$ dt around x = 0 has the form $f(x) = a_0 + a_1x + a_2x^2 + ...$

The coefficient a_2 (correct to two decimal places) is equal to ____.

$a_2 \longrightarrow 0$

$f(x) = \int_0^x e^{-t^2/2}$

$f(0) = \int_0^0 e^{-t^2/2} = 0$

$f''(0) = e^{-x^2/2} \cdot x$

$= 0$

$f'(x) = e^{-\frac{x^2}{2}} \frac{d}{dx}(x)$

$= e^{-\frac{x^2}{2}}$

$f'(0) = e^0 = 1$



Topic : Single Variable Calculus

#Q. Let the straight line x = b divide the area enclosed by $y = (1 - x)^2$, y = 0 and x = 0 into two parts R_1 $(0 \le x \le b)$ and R_2 $(b \le x \le 1)$ such that $R_1 - R_2 = \frac{1}{4}$. Then b equals

brain Storming

H.W

$b = \frac{1}{2}$

A 3/4

B 1/2

C 1/3

D 1/4



Topic : Single Variable Calculus

H.W

#Q. Let $f : [-1, 2] \to (0, \infty)$ be a continuous function such that $f(x) = f(1 - x)$ for all $x \in [-1, 2]$. Let $R_1 = \int_{-1}^{2} xf(x)dx$ and R_2 be the area of the region bounded by $y = f(x)$, $x = -1$, $x = 2$ and the x-axis. Then

A $R_1 = 2R_2$

B $R_1 = 3R_2$

C $2R_1 = R_2$

D $3R_1 = R_2$



2 mins Summary

Topic One — Area

Topic Two — Taylor Series

Topic Three

Topic Four

Topic Five

THANK - YOU